🔧 Non-Functional Requirements

Week 3 • Day 3 • Requirements Engineering & Mini-Capstone

TPM Academy

🎯 Day 3 Objectives

- Write NFRs in **Requirement / Defense / Verification** form — specific, defended, testable
- Cover all five categories: **Performance, Security, Accessibility, Observability, Compliance**
- Tie observability NFRs to the **Tier Sheet operational signals** from Week 2
- Surface **NFR trade-offs** explicitly — mature Product Requirements Documents (PRDs) name the tensions
- Deliver Section 7 with **6–10 NFRs** + a Known Trade-offs subsection

🕑 Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Recap Section 6; restate non-AI; print NFR template
09:15 – 10:30	Block 1 + Activity 1	NFR Triage
10:45 – 12:00	Block 2 + Activity 2	Performance + Observability
13:00 – 14:15	Block 3 + Activity 3	Security + A11y + Compliance
14:30 – 16:00	Block 4 + Activity 4 + Wrap	NFR Cross-Review + Trade-offs

Block 1 — What an NFR Is

Why most NFR sections fail

📚 The NFR Shape

An NFR is a **constraint** the system must hold under specified conditions, with a **measurable target**, a **defended rationale**, and a **verification method**.

```
### NFR — <short name="">
**Category:** Performance / Security / Accessibility / Observability / Compliance
**Requirement:** <specific, measurable="" target="">
**Defense:** <why this="" number;="" what="" scenario="" justifies="" it="">
**Verification:** <how we="" test="" observe="" in="" production=""></how></why></specific,></short>
```

Defense is the section that distinguishes TPM from boilerplate. "p95 < 400ms because dispatchers tap this 40 times per shift" beats "should be fast."

⚠️ The 5 NFR Failure Modes

Failure	Example
Boilerplate	"Performance must be acceptable"
Unmeasurable	"The system shall be highly available"
No defense	"p95 latency < 100ms" (with no rationale)
No verification	"The system shall be secure" (how would we know?)
Wrong category	A "Performance" NFR that's actually a feature requirement

📚 The Five Categories

Accessibility follows the **Web Content Accessibility Guidelines (WCAG)** 2.1 AA standard.

Performance

- Latency (p50 / p95 / p99)
- Throughput
- Resource caps
- Cold-start

Security

- AuthN / AuthZ
- Data classification
- Encryption
- Audit logging

Accessibility

- WCAG 2.1 AA
- Keyboard reachability
- Screen reader
- Reduced motion

Observability

- Logs (structured)
- Metrics (Tier Sheet!)
- Traces
- Alerts

Compliance

- Data residency
- Retention windows
- Industry-specific
- Audit trails

(Coverage)

- Each gets ≥ 1 NFR
- 6–10 total
- No category empty

👥 Activity 1 — NFR Triage

Triads • 35 minutes

10 NFR examples. Identify the failure mode(s) for each. Rewrite the 5 worst.

- 15 min: triage all 10
- 15 min: rewrite the 5 worst
- 5 min: identify the hardest one to defend even after rewriting

⏱️ 15 triage · 15 rewrite · 5 reflect

Block 2 — Performance + Observability

"What should we expect" + "How would we know"

⚡ Worked Performance NFR

```
### NFR — Reconcile flow latency
**Category:** Performance
**Requirement:** Modal opens within 1 second (p95) on the median
dispatcher device (mid-tier Android, 2-yr-old hardware)
on a 4G LTE connection.
**Defense:** Dispatchers tap "Reconcile all" 30+ times per shift.
At p95 = 1s, cumulative wait ≤ 30s/shift. At p95 = 3s,
wait = 90s/shift, crossing the threshold dispatchers
reported as "I'd rather use paper."
**Verification:** Synthetic transaction in Datadog from a fleet of
mid-tier devices. p95 reported daily; alert if 7-day
trailing p95 > 1.2s.
```

🔎 Worked Observability NFR

```
### NFR — Reconcile flow event coverage
**Category:** Observability
**Requirement:** The reconcile flow emits structured events for:
open_modal, ticket_select, ticket_deselect,
submit_attempt, submit_success, submit_failure (with
reason), abandon (modal closed without submit).
**Defense:** Three Tier-Sheet operational signals (abandonment rate,
submit-failure reasons, time-to-submit) require these
events. Without them, we cannot measure success.
**Verification:** Smoke test in staging asserts each event fires;
production dashboard tracks event counts per dispatcher
per shift.
```

The cross-check: every Tier Sheet operational signal needs an observability NFR enabling it. If you can't measure it, you can't move it.

👥 Activity 2 — Perf + Obs NFRs

Triads • 40 minutes

Each member proposes 1 Performance NFR; triad picks 2–3. Same drill for Observability.

- 15 min: 2–3 Performance NFRs with defenses
- 15 min: 2–3 Observability NFRs (reference Tier Sheet)
- 10 min: cross-check perf ↔ obs coverage

⏱️ 15 + 15 + 10

Block 3 — Security + A11y + Compliance

The "draft the first version" categories

🔒 Security NFRs — the standard set

1. **AuthN:** How do we know who the user is?
2. **AuthZ:** What can this user do? (Explicit not-allowed list)
3. **Data classification:** personally identifiable information (PII) / Payment Card Industry (PCI) / internal / public — what touches what?
4. **Encryption:** In transit via Transport Layer Security (TLS 1.2+); at rest (managed keys?)
5. **Audit:** What user actions are logged? Where? Retained how long?

> **The TPM job:** not to be the security expert. To draft the first version so the security team has a starting point.

♿ Accessibility NFRs (pull from Week 2 Day 3)

```
### NFR — Accessibility floor
**Category:** Accessibility
**Requirement:** The reconcile flow conforms to WCAG 2.1 AA. All
interactive elements keyboard-focusable; visible focus
indicators; 4.5:1 text contrast minimum; form fields
labeled; error messages identify the field.
**Defense:** Customer base includes dispatchers using assistive
technology (Interview 3: screen-reader user). ADA
exposure is non-trivial in our segment.
**Verification:** axe-core scan on every PR; manual screen-reader
pass before launch.
```

📝 Compliance NFRs — FieldPulse-flavored

```
### NFR — Audit trail retention
**Category:** Compliance
**Requirement:** Reconcile-flow events are retained for 24 months
in the audit event store, queryable by dispatcher_id
and date range.
**Defense:** SOC 2 Type II requires user-action audit trail;
dispatch decisions are subject to wage-and-hour audit
by some state authorities.
**Verification:** Quarterly audit-trail test: random user-event
from 18 months ago must be retrievable in <10 minutes.
```

If you have no compliance NFR, ask: "If a regulator audited this feature, what would they ask?" That answer is your NFR.

👥 Activity 3 — Sec + A11y + Comp

Triads • 40 minutes

3–5 Security NFRs (5-question set). 1–2 A11y NFRs (Week 2 Day 3). 1–2 Compliance NFRs.

- 15 min: Security (3–5 NFRs)
- 10 min: Accessibility (1–2 NFRs)
- 10 min: Compliance (1–2 NFRs)
- 5 min: defenses are real, not boilerplate

⏱️ 15 + 10 + 10 + 5

Block 4 — Trade-Offs & Cross-Review

Mature NFR sections name the tensions

⚔️ Four Classic NFR Trade-offs

Trade-off	The tension
Performance vs Observability	Heavy logging slows the hot path; light logging blinds you
Security vs Time-to-first-value	Strict authZ blocks the new-user happy path
Compliance vs Latency	EU data residency adds cross-region hops
Accessibility vs Visual Polish	Some animations conflict with reduced-motion / screen-reader

Mature sections: name the trade-off, name the resolution (or note the open tension).

📝 Worked Known-Trade-offs Subsection

```
### Known trade-offs (in this section)
```

```
<li><p><strong>Audit logging vs latency:</strong> NFR-7 (full event logging) adds
~30ms to the hot path tested in NFR-1. We accept this; if
hot-path latency becomes the bottleneck, we will move logging
to async dispatch.</p>
</li>
```

- ```
  Strict RBAC vs onboarding: NFR-10 (managers cannot view
  dispatcher reconciles in their first week) blocks an onboarding
  scenario. Resolved: managers see read-only view in week 1; full
  view in week 2.
  ```
  ```

👥 Activity 4 — NFR Cross-Review + Trade-offs

Triad pairs • 45 minutes • Wrap

Cross-review the NFR section (same pairing as Day 2). Surface trade-offs. Add the Known Trade-offs subsection.

- 10 min: read NFR section, check defenses + verifications
- 10 min: reviewer surfaces trade-offs
- 10 min: author triad responds
- 15 min: revise + add Known Trade-offs subsection

⏱️ 10 + 10 + 10 + 15 · 15-min wrap-share

🎉 Day 3 Wrap

What we built

- NFR Triage discipline (5 failure modes)
- Coverage across 5 categories
- Tier-Sheet-linked observability
- Known Trade-offs subsection

What opens tomorrow

- **Mini-capstone Product Requirements Document (PRD) assembly**
- Add Section 8 Metrics, Section 9 Risks, Section 10 Deps, Section 11 Out-of-scope
- Final pass + integration check
- Still non-AI

Bring to Day 4: Sections 1–7 complete. Tomorrow you ship.

💬 Questions & Reflection

Which of your NFRs would a senior engineer push back on hardest?